

Technology Stack

Date	14 July 2026
Team ID	XXXXXX
Project Name	Personalized Networking Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	Streamlit	Enables rapid, Python-native development of an interactive UI (bio/event input, generate button, chat-style output) without needing a separate JavaScript framework.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	FastAPI (Python)	Provides high-performance async REST endpoints, automatic Swagger/OpenAPI docs, and integrates cleanly with Hugging Face Transformers pipelines.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	Local JSON files (SQLite/PostgreSQL-ready)	Lightweight persistence (history.json, feedback.json) suited to a local single-user MVP, with a clear upgrade path to SQLite/PostgreSQL for multi-user scaling.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	Local deployment (localhost)	The project scope targets local demonstration and testing (Streamlit on :8501, FastAPI on :8000) rather than production cloud hosting.
5	Version Control & CI/CD	Git & GitHub	Supports source control and collaboration between team members, with versioned tracking of backend and frontend changes.

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6	Third-Party APIs / Other Tools (e.g., Stripe, Twilio, Google Maps)	Hugging Face Transformers (DistilBERT, GPT-2); Wikipedia API (Python wrapper)	DistilBERT gives fast zero-shot theme classification, GPT-2 generates natural conversation starters, and the Wikipedia API provides quick, reliable fact verification.